

Simulating Participant Behavior in Experience Sampling Method Research

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ABSTRACT

The Experience Sampling Method (ESM) is applied widely for collecting self-reports from participants in free-living environments. Preserving high compliance in ESM remains challenging, especially when a study lasts more than a few weeks. Markedly, participants get increasingly bothered by prompts delivered at inconvenient moments. To alleviate that, personalization techniques have shown their potential. Particularly, ESM protocols that delivered prompts at more convenient times have significantly fewer drop-outs. Such personalization may lead to sampling bias, while ESM should be ecologically valid. Therefore, it is critical to equip experimenters with tools that enable trade-off analyses between the minimization of dropout versus the maximization of ecological validity. This paper lays the foundations for such analyses: we propose a novel ESM-specific participant behavior simulator, demonstrate its resemblance to real-life data and expected behaviors indicated by psychological theories. Such simulators enable trade-off analyses and they can help avoid the cold start of reinforcement learning agents.

CCS CONCEPTS

• **Computing methodologies** → **Simulation types and techniques**; **Simulation tools**; *Learning settings*; • **Human-centered computing** → **HCI design and evaluation methods**.

KEYWORDS

experience sampling method, human behavior simulation, decision-making process

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1 INTRODUCTION

The experience sampling method (ESM) is a popular way to research people's behaviors, thoughts, and feelings during their daily activities with high validity and reliability [6, 18, 27]. ESM systematically spreads notifications throughout the day, asks study participants to self-report as soon as a notification is received, and usually lasts for a few weeks [27]. This method has been widely adopted in the field of Human-Computer Interaction (HCI) [5, 23], psychology [43], healthcare [27], and education [2]. Although ESM has overcome the flaws of its predecessors (i.e., observation and diary) mainly by reducing retrospection, it has limitations of its own. One of the main and well-documented challenges of ESM is the decline of response rate which is often times coupled with the dropout of the study participants. Dropout, especially if the study period is longer than a few weeks, can happen due to several reasons. For example, response fatigue [34, 42], changes in participant's motivation [47], type of ESM device [19], level of attachment to or the importance of the outcome of the study [27, 31], study design choices such as the number of notifications per day [10, 48], and receiving such notifications at inopportune moments [14]. Previous research has shown that finding opportune moments of delivering ESM notifications (also known as prompts or beeps) through personalization can help mitigate the hurdles mentioned above, potentially leading to longer retention periods and higher response rates [24, 25, 38].

More recently, reinforcement learning (RL) is gaining popularity as a method to adapt the timing of notifications based on implicit or explicit feedback from the participants [51]. Specifically, in the domain of health and well-being, adapting the time and content of intervention-related prompts has been highly effective [28, 33, 49, 50]. Even though adaptation and personalization contribute to finding opportune moments, it is a potential source of sampling bias. ESM, is intended to be ecologically valid [20]. That means data should be captured from various moments in one's life rather than only from those specific moments classified as opportune through participants' feedback. So, even though adaptation and personalization can improve interventions they might jeopardize the ecological validity of ESM studies as they may cause response bias [13]. We believe that there is a trade-off between ecological validity and the degree of personalization. When personalizing, ecological validity needs to be maintained by capturing data at moments left out as the result of adaptation. Therefore, the degree of personalization needs to be determined in such a way that ecological validity, retention, and response rate remain within an acceptable threshold. This trade-off is not yet sufficiently understood, calling for further research.

A further complication regarding the use of RL for personalizing the scheduling strategy in ESM is that when applying agent-based RL to find opportune moments, the learning agents usually start without any prior knowledge about when opportune moments are. They require multiple interactions with the participants to draw inferences and ultimately make effective adaptations. Such a problem is referred to as the cold start [44]. Cold start is a threat to the compliance of participants since at the beginning of the learning process many notifications may come across as intrusive. This results in discomfort for ESM study participants as they are already required to invest considerable time and commitment for self-reporting several times a day [11]. Currently, ESM study designers define their scheduling strategy based on insights from the study population or general heuristics for optimizing ESM scheduling, e.g., regarding the minimum separation between sampling moments, the times of the day where a high response rate can be expected, etc. Rather than programming such knowledge as scheduling rules, a potentially more scalable approach for countering cold start RL-related problems is to pre-train RL agents with virtual participants. Subsequently, when the actual study begins, agents will be less intrusive because they require fewer interactions as they have been pre-trained (warm start) [40, 52]. The virtual participant should be able to answer “having received a notification at a given moment, will the participant self-report considering the current state of mind and context?”. Such simulation has been argued to be theoretically possible by utilizing cognitive models [52]. Moreover, some psychological theories have practically been adapted to simulate the respondent’s decision-making process [50]. Building on these premises, our aim is to enable ESM researchers (experimenters) to easily use psychological theories, declare their own heuristics, apply trained machine learning models, and utilize historical data to create an interactive simulation environment with virtual participants driven by the aforementioned elements. In addition, we picture a collaborative platform where such components can be shared and reused by other experimenters as well. By doing so, we aspire to contribute to solving the scarcity of data needed to pre-train learning agents and to support experimenters in their study design choices.

In this paper, we focus on an ESM-specific simulation environment aiming to discover the opportune moments of interruption and demonstrate the necessity of incorporating notification *salience* (i.e., the amount of attention attracted [45]) to simulate realistically. The subsequent section overviews the relevant research on personalization, adaptation, and attempts to model the human mind congruent with the ESM context. In the methods section, we explain the psychological theories, heuristics, historical data, and machine learning models incorporated in our simulation. In the results section, we disclose our latest achievements by comparing the performance of our simulator with different settings. In the future work section, we announce the next steps that we deem meaningful after our late-breaking results. Lastly, we conclude by digesting our achievements and concisely stating our arguments in the conclusion section.

2 BACKGROUND

Decision-making is the cognitive process that results in the selection of a belief or execution of a task among several possible

alternative options. It is a reasoning process based on the values, preferences, and beliefs of the decision maker [41]. In the current paper, we aim to estimate whether a participant would respond to an ESM notification through simulation. Thus, the key action to be determined by the decision-making process of a virtual participant is either to respond/neglect a notification. Decision-making models are usually simplifications that highlight the domain specification [35]. Specifically, in the HCI domain, dynamic Bayesian networks (DBNs) are a common choice to model what affects decisions [21, 50]. Such networks represent the dependencies and temporal relationships graphically [7].

In the context of ESM, notifications act as triggers to remind study participants to self-report on a specific topic or fill out a survey that fulfills the research goal of the experimenter [27]. Whereas in the intervention domain, a notification is responsible to deliver the right type or amount of support at the right time [33]. Despite such a difference, the overall approach to personalizing the moment of delivery is quite similar. For example, the *cost* of interruption has been well-studied in the field to facilitate learning predictive models that can be utilized to personalize the timing of interruption [21, 22, 24]. Besides, *context*, as in user’s activity, location, time of day, emotions, and engagement has also been incorporated to identify suitable moments of interruption [38]. Usually, in addition to sensor data, variables such as the response time and the number of notifications opened help in driving adaptations [37].

As mentioned earlier, other than adaptation and personalization on the fly, simulation can be adopted to avoid high costs to users, specifically when adaptation is backed by machine learning methods such as RL [22, 46]. To imitate how the human mind functions, various cognitive architectures and general computational frameworks such as ACT-R [1], Soar [26], and BOID [3] have been used. Generally, applying a conceptual cognitive model to a specific use case and deriving a computational representation is challenging. For example, in the context of ESM, the Fogg behavior model seems to be highly compatible as it already contains the notion of triggers (notifications) [12]. However, it cannot be easily applied in the simulation because sufficient quantitative support is not provided [4, 17]. Moreover, the attempts for its formalization remain highly heuristics-based [15]. Accordingly, we envision a platform where applicable computational models can not only be used individually by researchers but also shared and then reused by other scholars. Eventually, the quality of such models can be assessed based on their performance, reusability, and popularity. Such a simulation environment could encourage the formalization of other highly conceptual and prevalent theories such as COM-B [32] and SDT [8].

In this paper, for our ESM participant behavior simulation, we adopt the derived psychological models of *memory decay* (i.e., the forgetting curve [39, 45]) as well as *urge* that have been previously used for simulation in the context of a health intervention [50]. Additionally, as momentary *context* affects decision-making, it needs to be integrated with the simulation environment as well [9, 25].

3 METHODS

In this section, we describe the psychological theories, heuristics, and trained machine learning models we use to simulate the human

decision-making process. Additionally, we introduce the data set that we used to validate the performance of our simulation alongside the parameters and components of our ESM-specific simulation environment.

3.1 Theories, Heuristics, and Machine Learning

We model the human decision-making process and the environment as discrete-time stochastic processes. Internal processes such as human memory are modeled as functions of time and interruptable via events (such as receiving a beep). Such a modeling problem is commonly formalized as a Markov Decision Process (MDP) [44]. Accordingly, to model our virtual participant, it is required to identify the dependencies in the decision-making process. To solidify our beliefs and determinants of decision-making in the ESM context, we constructed a Bayesian network. Fig. 1 illustrates the variables and conditional dependencies of our provisional Bayesian network including our previous works on personalization [30] and physical activity [25] that assess their impact on compliance in the ESM context. We discovered that the *action* (i.e., respond to/ignore a notification), *personal context* (e.g., user's activity, location, and emotions), and *time* are common components in constructing the decision-making process model [21, 45, 50]. To conveniently arrange the determinants, we abstract all the variables by the COM-B model (adjacent nodes of *action*). In this paper, we only adopt computational models of *urge* (motivation), *memory accessibility*, *beep*, *salience*, and *personal context* to drive our baseline ESM-specific simulator to estimate *action*.

We extended our Bayesian network into a dynamic Bayesian network (DBN) to identify the temporal relationships between states. Thereby, we propose the following transition probabilities (inspired by the equations presented by Wang et al. [50]) between states and time steps to construct our MDP: At the beginning of the simulation process (t_0), our virtual participant has a baseline motivation (M_{t_0}) to respond to ESM beeps. If an ESM beep is received ($B_t = 1$) and responded to ($A_t = 1$), the motivation to react immediately again should drop, then increase gradually. So, M_t drops to zero and increases linearly over time at the rate of μ (Eq. 1) based on the previous action (A_{t-1}), level of motivation (M_{t-1}) and α_t . Through α_t we assume that when beeps are received at inopportune moments, the motivation of the study participants declines over time (Eq. 4; c_α is a constant). As soon as a beep is received, the memory accessibility (MA_t) is increased to the maximum possible value and it decays exponentially over time at the rate of σ (Eq. 2). Accordingly, with β_t we assume that when a similar beep (same text, notification sound, vibration pattern, etc.) is received recurrently, it becomes less noticeable, hence, its salience should decrease (Eq. 5; c_β and b_β are constants) and diminish the maximum value of memory accessibility (β_t in Eq. 2). In addition, We consider C_t (momentary context) as the physical activity type captured by the smartwatch where $P(C_t|A_t)$ and $P(C_t)$ are defined as the probability of a certain physical activity (recognized by the activity monitor of the smartwatch) given an action (respond/neglect) and probability of a given physical activity, respectively. Ultimately, considering Markov property (i.e., the future state (s_{t+1}) is only dependent on the current state (s_t)) and contextual probabilities ($P(C_t|A_t)$, and $P(C_t)$) derived from historical data (see subsection 3.2), the probability of responding to a beep (A_t), following Bayes theorem, can

be calculated by Eq. 6 [49]. To solve the right-hand side of the aforementioned equation, calculating $P(A_t|MA_t, M_t)$ is relaxed by assuming Eq. 3.

$$P(M_t|A_{t-1}, M_{t-1}) = \begin{cases} 0 & \text{if } A_{t-1} = 1, \\ \min\{M_{t-1} + \mu, 1 + \alpha_t\} & \text{if } A_{t-1} = 0. \end{cases} \quad (1)$$

$$P(MA_t|B_t, MA_{t-1}) = \begin{cases} \min\{1, 1 + \beta_t\} & \text{if } B_t = 1, \\ MA_{t-1} \cdot \sigma & \text{if } B_t = 0. \end{cases} \quad (2)$$

$$P(A_t|MA_t, M_t) = MA_t \cdot M_t \quad (3)$$

$$\alpha_t = \begin{cases} \min\{1, \alpha_t + c_\alpha\} & \text{if } B_t \equiv \text{opportune}, \\ \alpha_t - c_\alpha & \text{otherwise}. \end{cases} \quad (4)$$

$$\beta_t = \begin{cases} \beta_t - c_\beta & \text{if } B_t = 1, \\ \min\{1, \beta_t + b_\beta\}; b_\beta \ll c_\beta & \text{otherwise}. \end{cases} \quad (5)$$

$$P(A_t|C_t, B_t, A_{t-1}, MA_{t-1}, M_{t-1}) = \sum_{MA_t} \sum_{M_t} \frac{P(C_t|A_t)}{P(C_t)} \times P(A_t|MA_t, M_t) \times P(MA_t|B_t, MA_{t-1}) \times P(M_t|A_{t-1}, M_{t-1}) \quad (6)$$

3.2 Data

To realize the context-related probabilities ($P(C_t|A_t)$, and $P(C_t)$) we make use of our data set of a former ESM study that we carried out with $N = 71$ participants for a period of 4 weeks [25, 36]. We reused the exact historical data to create the context of virtual participants. To estimate $P(C_t|A_t)$, in our interaction log data, we labeled each moment of beep delivery as opportune when the participant responded to that beep and inopportune when no corresponding response was recorded. Then we used the data to train a probabilistic decision tree classifier to estimate the aforementioned probability [16]. Ultimately, we aim to make such parameters, their values, and the outcome of the simulation, as well as trained machine learning models and (anonymized) study data shareable and reusable such that ESM researchers can benefit and build on top of previous experiments in a low-code/no-code environment, which is out of the scope of the current paper.

3.3 Simulation

Discrete event and agent-based simulations are commonly used to model human behavior [29]. Agent-based methods usually involve the interaction between people (agents) whereas, in the discrete event simulation, the focus is on modeling processes. As we aim to model the decision-making *process*, a discrete event approach is followed. The *environment* of our ESM-specific simulation is set up according to an ESM protocol. e.g., the study period (as in the number of days) determines how many times the simulator's clock should tick. Besides, the beeping *process* is also set up according to the beeping schedule of the ESM protocol and determines when the beeping process should *trigger* a beep *event* to *interrupt* the virtual participant. Concurrently, the decision-making process is updated by every tick and upon receiving each beep according to the formulas in subsection 3.1.

Our goal for simulation in this paper is to demonstrate that in hindsight, we could have chosen our study design parameters (e.g., the number of notifications per day) differently such that the retention would be prolonged or the response rates would increase. To compare the simulated responses with actual ones, we created two binary arrays, one for simulated interactions and another one

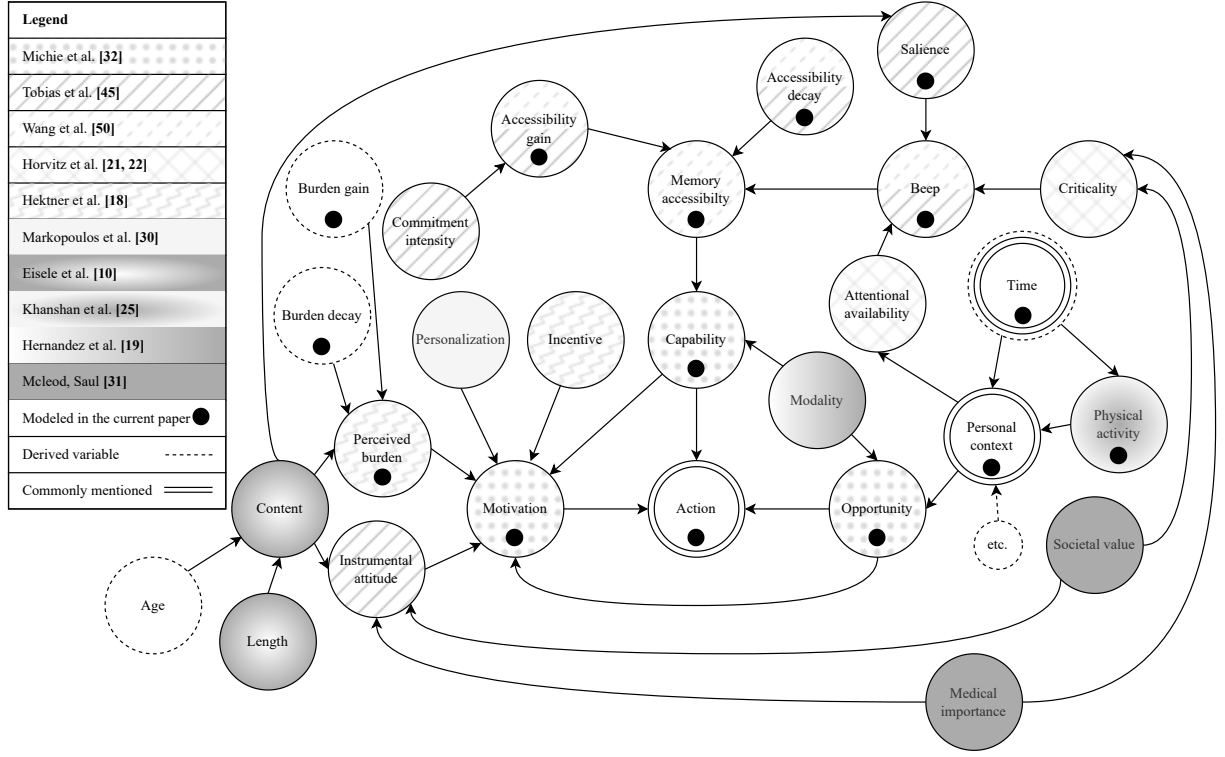


Figure 1: Our *provisional* dynamic Bayesian network that shows what variables could influence action in a single time step.

for actual ones. The number of items in each array corresponds to the number of beeps and the value of each element represents the action taken by the participant: 0 means “no response”, and 1 means “response”. Accordingly, to tune the parameters of our simulation so that it fits our interaction log data, we applied the grid search method with the Jaccard coefficient as its objective function. That coefficient is commonly used to compare the similarity of two objects (in our case: two binary arrays of interactions) [16].

4 RESULTS

In this section, we demonstrate that in hindsight, having a simulation could support making decisions about study design choices such as the number of notifications or the sampling policy (context-sensitive, random, or interval-based).

4.1 Setup

Initially, to fit the parameters of our simulation, we aimed at aligning not only the number of responses of our virtual participants with the real ones but also the overall response pattern and behavior. Hence, the Jaccard coefficient was used (see also subsection 3.3). As a result of the fitting, the assignments $\mu = 0.2$, $\sigma = -0.04$, $c_\alpha = 0.005$, $c_\beta = 0.007$, $b_\beta = 0.002$, $\alpha_{t_0} = 0$, and $\beta_{t_0} = 0$ were made. Then, as we intended to investigate how well our simulator imitates real life, we set the simulation duration for each virtual participant exactly mapped with the real participants in our historical data (see 3.2). In addition, we instructed the beeping process to send beeps at the same moments as the actual study (original policy). Moreover, the

decision-making process of the virtual participants is designed as stochastic. Thus, each run yields slightly different outputs.

4.2 Response rate

To analyze the performance of our simulator, we take into account the response rate. i.e., the ratio of responses to the total beeps. We expect that the performance of our simulator would be similar to the real study in terms of the root mean squared error (RMSE) over response rates. We fit the parameters according to the Jaccard coefficient rather than the RMSE of response rate because, in our experience, the interaction of the virtual participants did not resemble real-life behavior when RMSE was chosen as the optimization criteria. Additionally, we included response count in our analysis as studies with only a few beeps may get high response rates but still low overall response counts, which might be a more important factor for some studies. To highlight the different sampling regimes, in addition to running our simulation with the same sampling regime as the real study (original policy), we run our simulation for 12 more scenarios. In that, we explore context-sensitive, interval-based, and random policies each with 4 variants: Less intrusive ($\approx 50\%$ less number of beeps than the original policy), much less intrusive ($\approx -75\%$), more intrusive ($\approx +25\%$), and much more intrusive ($\approx +50\%$). We keep the same parameters for the decision-making process of our virtual participants ($\mu = 0.2$, $\sigma = -0.04$, $c_\alpha = 0.005$, $c_\beta = 0.007$, $b_\beta = 0.002$, $\alpha_{t_0} = 0$, $\beta_{t_0} = 0$) for each scenario. Subsequently, we anticipate that the variants with lower intrusiveness result in higher response rates than those with higher intrusiveness. It is worth noting that the original policy

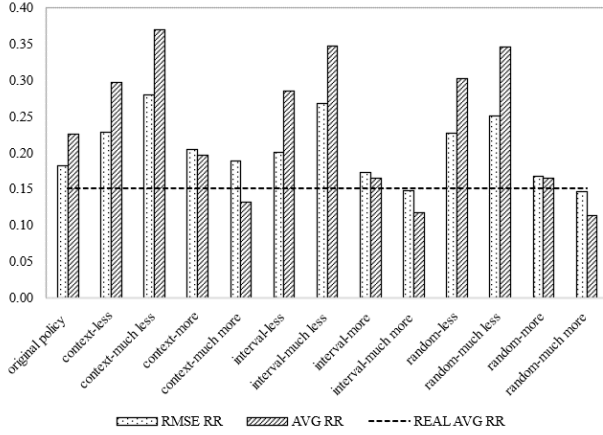


Figure 2: RMSE of simulated vs. real response rate (RMSE RR), average response rate (AVG RR), and the average response rate of the real study (REAL AVG RR)

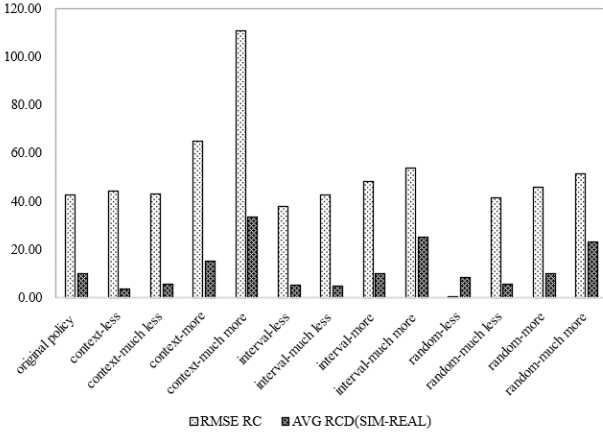


Figure 3: RMSE of simulated vs. real response count (RMSE RC), and the average response count difference (AVG RCD) between original policy and simulation

(based on the collected data of our former ESM study explained in 3.2) was context-sensitive with two groups, one group receiving notifications at higher levels of physical activity and the other group otherwise. In the context-sensitive variant of the simulation, the moment of delivery is assessed based on the decision tree trained with our historical data.

As anticipated, the results of our simulation show that the more intrusive a policy, the lower the response rate (see Fig. 2). Such results are congruent with equations in 3.1 since α_t in Eq. 4 lowers the power of regaining motivation when beeps arrive at inopportune moments and β_t in Eq. 5 reduces the salience of a beep if received repeatedly and similarly. It can also be observed that context-sensitive variants have the highest response rates overall. As a supplement, AVG RCD shows on average how many more responses could be obtained compared to the original policy. Accordingly, it can be noted that more intrusive approaches can not significantly improve the total reports compared to the original policy, except for the

highly intrusive variants (see Fig. 3). We fit and then assigned the parameters of our simulation based on the whole population rather than separately for each individual, besides, our data set included participants that received more beeps than intended for a period due to a technical bug, and we compared against response rate rather than Jaccard coefficient, so in principle, we expected that a few variants might outperform the original policy in terms of RMSE (e.g., more intrusive variants of random sampling). The source code of our simulator is available at GitHub.

5 FUTURE WORK

The findings above indicate the feasibility of simulating user response behavior, based on a small set of theoretical assumptions regarding their attention, memory, motivation, and context. More specifically, in Fig. 1, our proposed dynamic Bayesian network illustrates a set of relevant determinants for assessing compliance in the context of ESM. In particular, we modeled action, motivation, opportunity, and capability through the perceived burden, memory accessibility, personal context, beep, and salience. These models are relatively simple, or even simplistic, regarding the true nature of human cognition and affect, but already help produce reasonable approximations of user response behavior, that can guide choices regarding different sampling schedules.

Further work could aim to improve the accuracy of the simulation either by incorporating more advanced theories of human cognition or by training models on larger data sets (more users, and longer periods), especially data sets including richer contextual data. Our specific simulation could be further improved by modeling more precisely the response behavior of the user: the effort required, the interface used and the time such a response takes, and provide a more holistic measurement of compliance to include, next to the response rate, also the quality of the responses, the retention rate, ecological validity, and the quality and richness of the response. Furthermore, the simulator could be improved by providing reusable and shareable components that are accessible through an interactive interface to the experimenters. On another note, future work can entail empirical elaborations about cold start RL-related problems and to what extent they can be alleviated through ESM simulation.

6 CONCLUSION

In this paper, we argued that attaining sufficient response rates and preventing early dropouts during the Experience Sampling Method (ESM) studies can potentially be alleviated through personalization and adaptation techniques. The effectiveness of such techniques had already been shown in the intervention domain. However, reusing the very same techniques may result in sampling bias in the context of ESM. To address that, we showed that simulating user behavior, as in their response to the ESM notifications, by training upon real-life data of ESM studies as well as incorporating relevant computational psychological theories can produce data similar to actual participant behavior. Accordingly, we suggested a minimal set of theories including attention, memory, and motivation that can estimate the participant reactions. We also modeled the context of participants (a significant determinant of one's momentary decision) based on their momentary level of

physical activity. We demonstrated that in hindsight, such simulation can help support the decisions of study designers as it can estimate the impact of their decision (such as the number of ESM notifications scheduled to be sent every day) on the adherence of study participants. In addition, we discussed that machine learning techniques, such as reinforcement learning (RL), can interact with such simulated environments and be pre-trained for predicting the opportune moments of prompting participants. Although that remains to be investigated empirically. Lastly, to show the flexibility and generality of our model, we simulated different ESM scheduling techniques and showed the correspondence of simulation outcomes to our historical ESM data as well as our expectations derived from psychological theories of human behavior and decision-making.

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